

Study of SOC and SOH Estimation Methods in Lithium-Ion Battery Management Systems

Submitted by:

Shrijan Samrat

Summer Intern, NIT-C

GUIDE: Dr. Nimal Madhu

1.Introduction:

Lithium-ion batteries have become one of the most important energy storage technologies in modern electronic systems due to their high energy density, lightweight structure, long cycle life, and high efficiency. These batteries are widely used in electric vehicles (EVs), portable electronic devices, renewable energy storage systems, aerospace applications, and smart grid technologies. As the adoption of battery-powered systems continues to increase globally, efficient battery monitoring and management have become essential for ensuring battery safety, reliability, and performance.

Battery Management Systems (BMS) play a critical role in monitoring and controlling battery operation. Important battery parameters such as State of Charge (SOC), State of Health (SOH), temperature, voltage, and current must be continuously estimated to ensure safe and efficient battery utilization. Among these parameters, SOC estimation is one of the most important and challenging tasks because SOC cannot be directly measured using physical sensors. Instead, it must be estimated using mathematical models, electrical measurements, and intelligent estimation algorithms.

Accurate SOC and SOH estimation are essential for improving battery lifespan, preventing overcharging and deep discharging, optimizing charging strategies, and enhancing the operational reliability of electric vehicles and energy storage systems. However, battery behavior is highly nonlinear and depends on several factors such as battery chemistry,

temperature, aging, operating conditions, and load variations. Different lithium-ion chemistries such as Lithium Iron Phosphate (LFP), Nickel Manganese Cobalt (NMC), and Lithium Cobalt Oxide (LCO) exhibit different voltage characteristics and electrochemical behavior, further increasing estimation complexity.[1][2]

Over the years, several SOC and SOH estimation techniques have been developed, ranging from classical methods such as Coulomb Counting and Open Circuit Voltage (OCV) estimation to advanced Kalman Filter algorithms and modern machine learning approaches. Recently, hybrid estimation techniques combining physics-based models and artificial intelligence methods have gained significant research attention due to their improved estimation accuracy and robustness.

In addition, the emergence of TinyML and Edge-AI technologies has created new opportunities for implementing lightweight intelligent SOC estimation algorithms directly on embedded Battery Management Systems with limited computational resources. These technologies enable real-time battery monitoring, low-power operation, and efficient deployment of machine learning models on microcontrollers and edge devices.[1][2]

This report presents a detailed study of SOC and SOH estimation methods used in lithium-ion battery systems. Classical estimation methods, Kalman Filter algorithms, machine learning approaches, hybrid estimation techniques, research challenges, and TinyML-based future possibilities are analyzed and compared with respect to accuracy, computational complexity, real-time capability, and embedded implementation suitability.

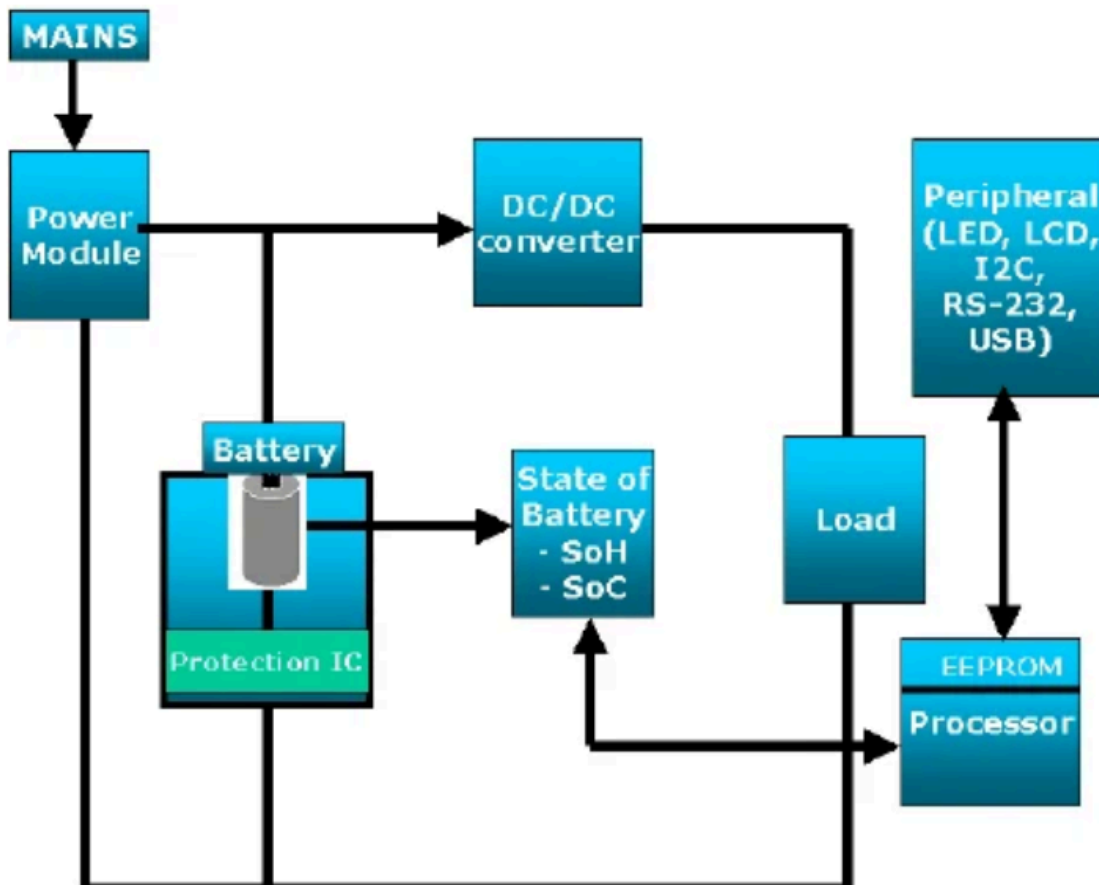


Figure 1: Basic Architecture of a Lithium-Ion Battery Management System

2. SOC and SOH Fundamentals 🪫 :

2.1 State of Charge (SOC) :

State of Charge (SOC) represents the remaining available capacity of a battery relative to its maximum capacity and is usually expressed as a percentage. SOC indicates how much charge is left inside the battery and is similar to the fuel level indicator in conventional vehicles. Accurate SOC estimation is essential for preventing overcharging and deep discharging, improving battery life, and ensuring reliable operation of Battery Management Systems.

SOC cannot be directly measured using physical sensors and must therefore be estimated using electrical measurements such as voltage, current, and temperature along with estimation algorithms and battery models. Accurate SOC estimation becomes difficult because lithium-ion battery behavior is highly nonlinear and varies with operating conditions, temperature, battery chemistry, and aging effects.[3][4]

SOC Equation

$$\text{SOC (\%)} = (Q_{\text{remaining}} / Q_{\text{rated}}) \times 100$$

Where:

- $Q_{\text{remaining}}$ = remaining battery charge
- Q_{rated} = rated battery capacity

2.2 State of Health (SOH):

State of Health (SOH) represents the overall health condition and aging status of a battery compared to its ideal condition when new. SOH is used to evaluate battery degradation, performance capability, and remaining operational effectiveness over time.

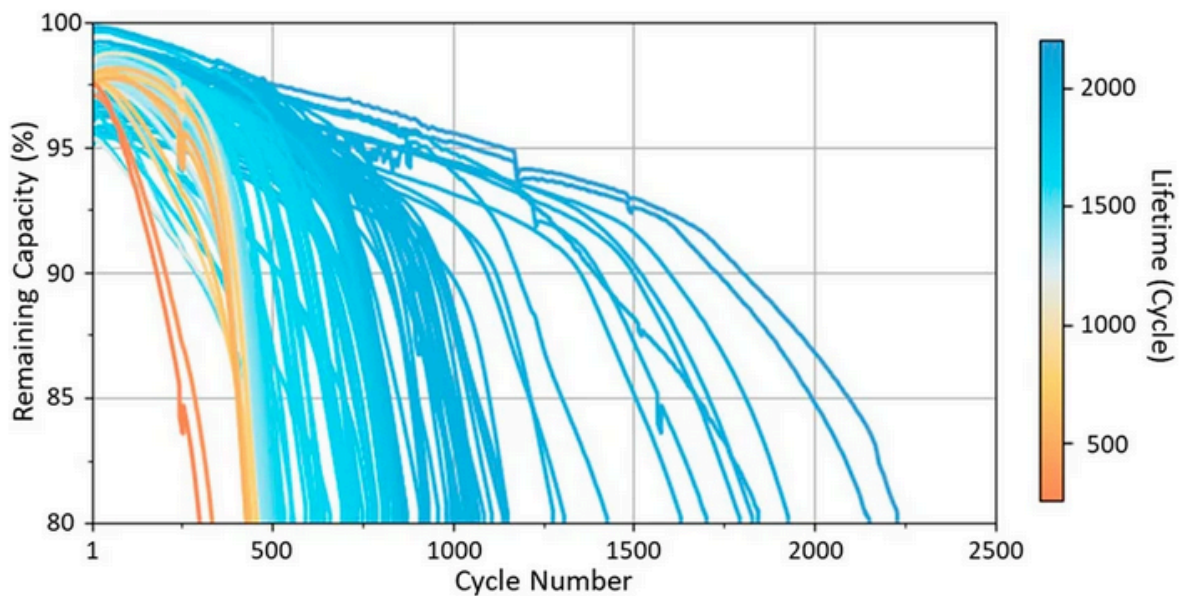
As lithium-ion batteries age, their maximum capacity gradually decreases and internal resistance increases due to electrochemical degradation and repeated charging/discharging cycles. SOH estimation is important for predicting battery reliability, scheduling maintenance, improving battery safety, and estimating remaining useful life in electric vehicles and energy storage systems.[3][4]

SOH Equation

$$\text{SOH (\%)} = (\text{Current Capacity} / \text{Rated Capacity}) \times 100$$

Where:

- Current Capacity = present battery capacity
- Rated Capacity = original battery capacity



Battery capacity fade curves of the dataset. The lifetime ranges from 300 to 2300 cycles.

Figure 3: Capacity Degradation and State of Health Variation in Lithium-Ion Batteries

2.3 Remaining Useful Life (RUL):

Remaining Useful Life (RUL) represents the estimated operational life remaining before a battery reaches its end-of-life condition. RUL prediction helps in maintenance planning, battery replacement scheduling, and improving the operational reliability of battery-powered systems.

RUL estimation depends on several factors such as battery aging, charging/discharging cycles, temperature conditions, and degradation behavior.

Accurate SOH estimation is often used as a basis for predicting battery RUL in modern Battery Management Systems.[5]

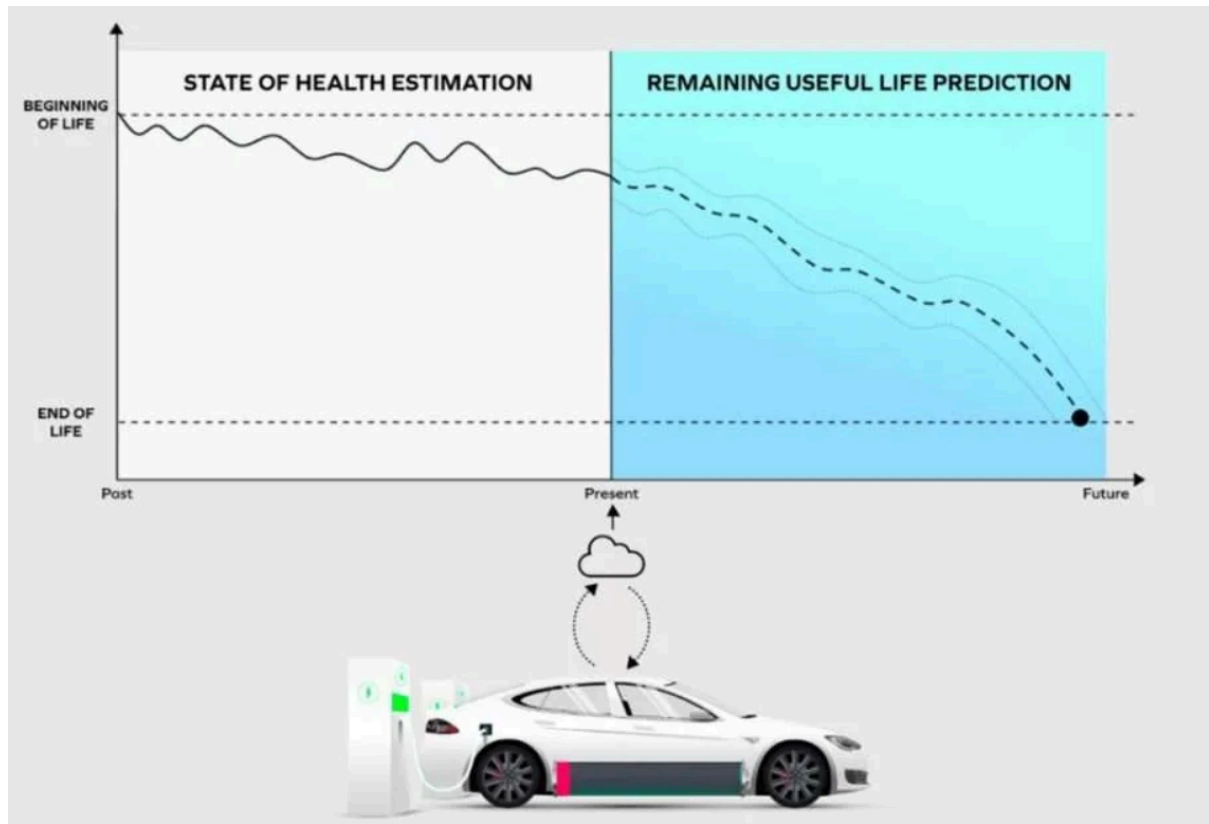


Figure 4: Remaining Useful Life Prediction of Lithium-Ion Batteries

2.4 Challenges in Battery State Estimation:

Battery state estimation is a challenging problem because lithium-ion battery behavior is highly nonlinear, dynamic, and chemistry-dependent. Parameters such as voltage, current, temperature, internal resistance, and aging continuously affect battery response during operation.

Different battery chemistries such as LFP, NMC, and LCO exhibit different voltage-SOC characteristics and thermal behavior, making universal estimation models difficult to develop. In addition, sensor noise, measurement uncertainty, environmental conditions, and battery degradation further increase estimation complexity.

Therefore, advanced estimation algorithms, machine learning techniques, and hybrid intelligent models are increasingly being explored to improve SOC and SOH estimation accuracy in modern Battery Management Systems.

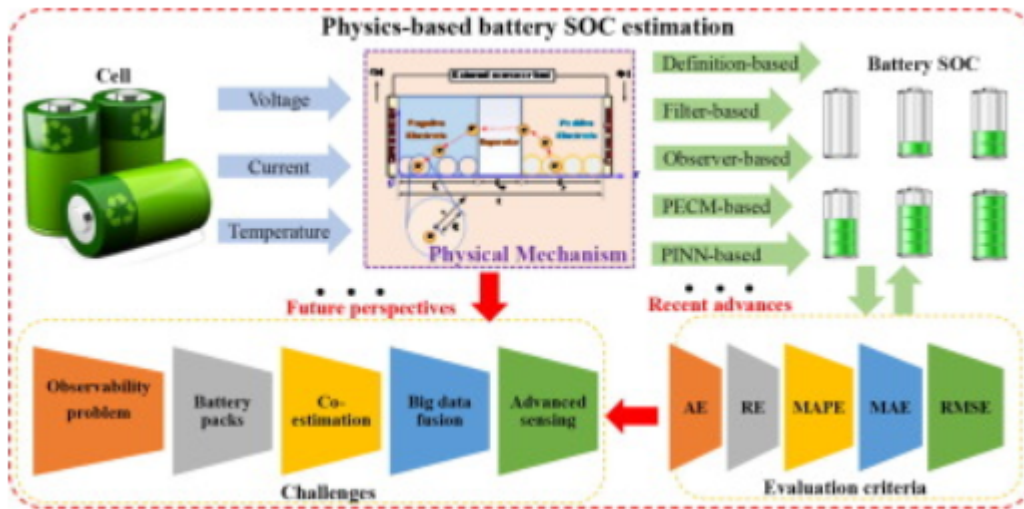


Figure 5: Factors Affecting Battery SOC and SOH Estimation Accuracy

3. Classical SoC Estimation Methods:

3.1 Coulomb Counting Method:

The Coulomb Counting method, also known as Ampere-Hour Integration, is one of the most widely used classical SOC estimation techniques. This method estimates battery SOC by continuously measuring the battery current and calculating the amount of charge entering or leaving the battery over time.

The basic principle of Coulomb Counting is based on current integration. During charging, current entering the battery increases SOC, whereas during discharging, current leaving the battery decreases SOC. By integrating the measured current over time, the remaining battery charge can be estimated.

The Coulomb Counting method is simple, computationally efficient, and suitable for real-time implementation in Battery Management Systems. However, the method strongly depends on accurate current sensing and an accurate initial SOC value. Small current measurement errors and sensor drift

accumulate over time, resulting in estimation inaccuracies during long-term operation.[4][5].

Equation:

$$\text{SOC}(t) = \text{SOC}(t_0) - (1/C_n) \int I(t) dt$$

Where:

- $\text{SOC}(t_0)$ = initial SOC
- C_n = nominal battery capacity
- $I(t)$ = battery current

Working Principle:

The battery current is continuously measured using current sensors. The measured current is integrated over time to calculate the amount of charge transferred during charging and discharging operations. The calculated charge variation is then used to update the battery SOC value continuously in real time.

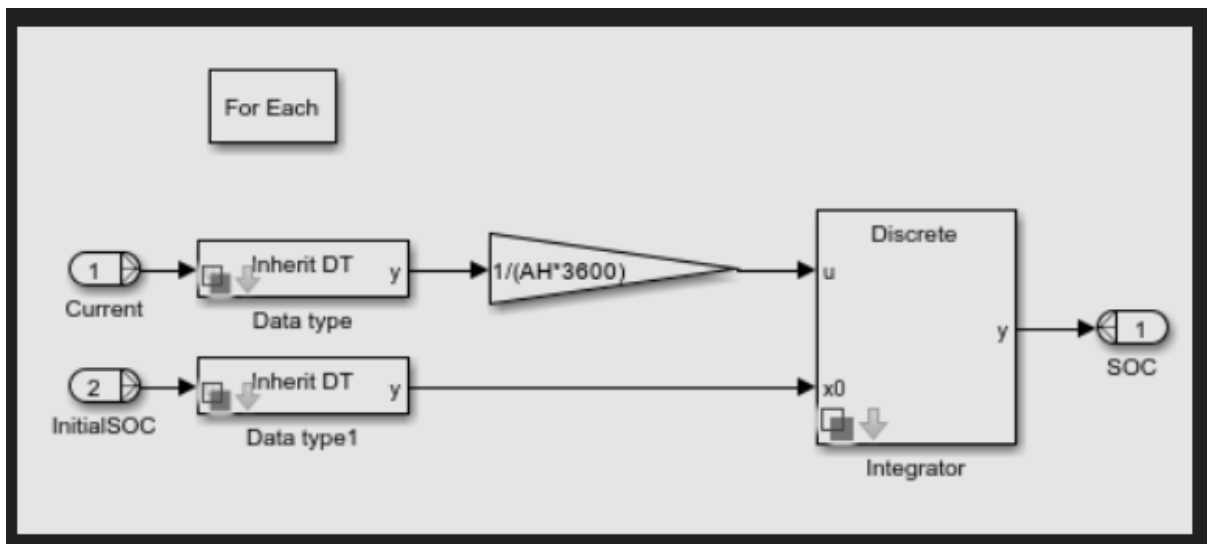


Figure 6: Block Diagram of Coulomb Counting Based SOC Estimation

Advantages:

- Simple implementation
- Low computational complexity

- Real-time estimation capability
- Widely used in BMS

Limitations

- Requires accurate initial SOC
- Sensor drift causes cumulative errors
- Error accumulation over time
- Reduced long-term estimation accuracy

Applications in BMS :

The Coulomb Counting method is widely used in Battery Management Systems for electric vehicles, portable electronics, and energy storage systems due to its simplicity and fast real-time implementation capability. It is often combined with advanced estimation algorithms such as Kalman Filters to improve long-term estimation accuracy.[4][5]

3.2 Open Circuit Voltage (OCV) Method :

The Open Circuit Voltage (OCV) method is another widely used classical SOC estimation technique in Battery Management Systems. This method estimates battery SOC by measuring the stabilized battery voltage under no-load resting conditions and mapping it to corresponding SOC values using OCV-SOC characteristics.

When a battery is actively charging or discharging, the terminal voltage is affected by internal resistance, current flow, and dynamic operating conditions. Therefore, the measured voltage may not accurately represent the true battery state. In the OCV method, the battery is allowed to rest for a certain period so that the voltage stabilizes. The stabilized voltage is referred to as Open Circuit Voltage (OCV).

Different SOC levels correspond to different OCV values. By comparing the measured OCV with pre-established OCV-SOC lookup tables or characteristic curves, the battery SOC can be estimated. The OCV method generally provides good estimation accuracy under resting conditions and is commonly used for battery calibration and initialization in Battery Management Systems.[3][4]

OCV-SOC Relationship :

The relationship between Open Circuit Voltage and State of Charge is highly dependent on battery chemistry. Different lithium-ion chemistries such as LFP, NMC, and LCO exhibit different OCV-SOC characteristics. In some batteries, voltage changes significantly with SOC variation, whereas in others the voltage remains relatively flat over a large SOC range.

OCV (V)	Approximate SOC (%)
4.20	100
4.00	80
3.80	60
3.60	40
3.40	20
3.00	0

Table 1: Example OCV-SOC Lookup Table for Lithium-Ion Batteries

Resting Voltage:

The OCV method requires the battery to remain under open-circuit or no-load conditions for a certain time period before voltage measurement. This resting period allows electrochemical stabilization inside the battery and reduces transient voltage effects caused by current flow and internal resistance.

LFP Battery OCV Curve :

Lithium Iron Phosphate (LFP) batteries exhibit a very flat OCV-SOC curve over a wide SOC range. Small voltage changes correspond to large SOC variations, making voltage-based SOC estimation difficult for LFP batteries.

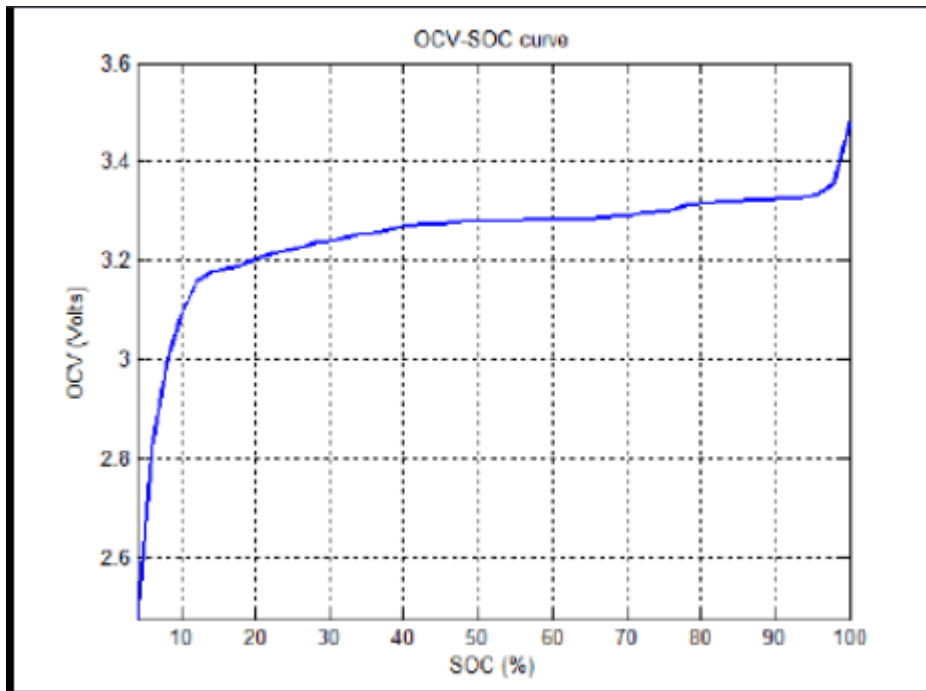


Figure 8: OCV-SOC Characteristic Curve of NMC Battery

LCO Battery OCV Curve:

Lithium Cobalt Oxide (LCO) batteries exhibit moderate nonlinear voltage variation with SOC and are widely used in portable electronic devices such as smartphones and laptops.

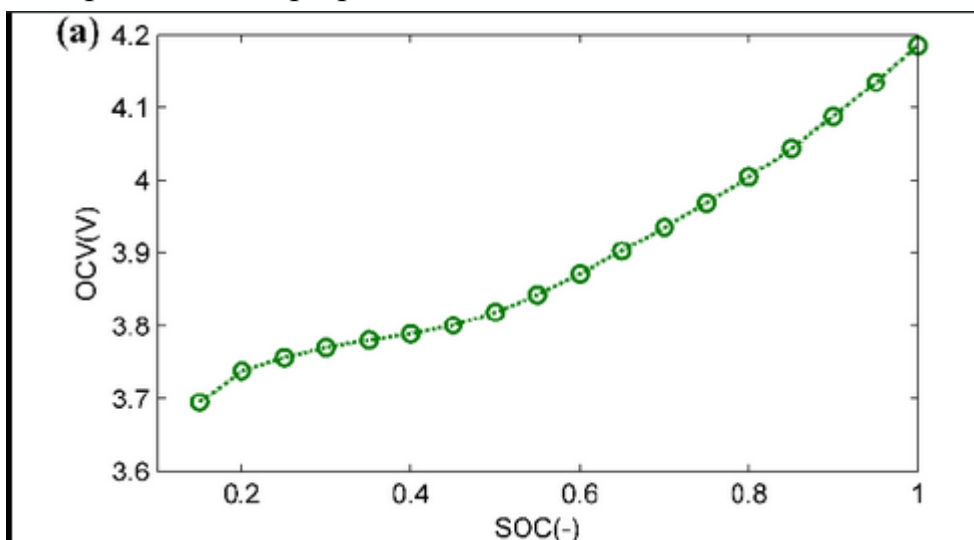


Figure 9: OCV-SOC Characteristic Curve of LCO Battery

Advantages

- Simple implementation
- Good estimation accuracy under resting conditions
- Useful for battery calibration
- Widely used in BMS initialization

Limitations

- Requires battery resting time
- Difficult for dynamic real-time estimation
- Strongly chemistry dependent
- Reduced accuracy during active load conditions

Accuracy Challenges :

The accuracy of the OCV method is affected by temperature variation, battery aging, chemistry differences, and incomplete voltage stabilization. Real-time applications such as electric vehicles rarely allow sufficient battery resting time, making standalone OCV estimation difficult under practical operating conditions.[3][4][6]

3.3 Comparison of Classical Methods :

Classical SOC estimation methods such as Coulomb Counting and Open Circuit Voltage estimation are widely used in Battery Management Systems because of their simplicity and low computational requirements. However, both methods exhibit different operational characteristics, advantages, and limitations under practical battery operating conditions.

The Coulomb Counting method provides continuous real-time SOC estimation by integrating battery current over time. It is computationally efficient and suitable for embedded implementation. However, the method suffers from cumulative error growth due to sensor drift and inaccurate initial SOC values.

In contrast, the OCV method provides comparatively accurate SOC estimation under resting conditions by utilizing stabilized battery voltage measurements. However, the method requires battery relaxation time and is not well suited for dynamic real-time applications such as electric vehicles where batteries continuously operate under load conditions.

Therefore, modern Battery Management Systems often combine classical estimation methods with advanced algorithms such as Kalman Filters and machine learning approaches to improve overall estimation accuracy and robustness.[4][5]

Parameter	Coulomb Counting	OCV Method
Principle	Current integration	Voltage-SOC relationship
Input Parameter	Current	Open Circuit Voltage
Real-Time Capability	High	Limited
Computational Complexity	Low	Low
Accuracy	Moderate	Good under resting conditions
Initial SOC Requirement	Required	Not always required
Error Accumulation	High	Low
Resting Requirement	No	Yes
Chemistry Dependence	Moderate	High
Main Limitation	Sensor drift and cumulative error	Requires stabilization time
Common Applications	Real-time BMS estimation	Calibration and initialization

Table 2: Comparison of Classical SOC Estimation Methods

Although classical SOC estimation methods are simple and computationally efficient, their limitations under nonlinear and dynamic battery operating conditions reduce long-term estimation accuracy. These limitations motivated the development of advanced estimation algorithms such as Kalman Filters and intelligent machine learning-based SOC estimation techniques.[4][5][6]

4. Kalman Filter Methods 🤖 :

Kalman Filter-based estimation techniques are among the most widely used advanced SOC estimation methods in modern Battery Management Systems. These methods improve estimation accuracy by combining mathematical battery models with real-time sensor measurements while handling noise and uncertainty in the system.

Unlike classical estimation methods, Kalman Filter algorithms continuously perform prediction and correction operations to estimate battery SOC dynamically. These algorithms are capable of reducing estimation errors caused by sensor noise, model uncertainty, and current drift effects.

Kalman Filter algorithms are especially important for lithium-ion batteries because battery behavior is highly nonlinear and varies with temperature, aging, and operating conditions. Different variants of Kalman Filters such as the Basic Kalman Filter (KF), Extended Kalman Filter (EKF), and Unscented Kalman Filter (UKF) have been developed to improve estimation performance for nonlinear battery systems.[4][7]

4.1 Basic Kalman Filter (KF) ⚡ 🤖 :

The Basic Kalman Filter (KF) is a recursive state estimation algorithm used to estimate unknown system states using mathematical prediction and measurement correction techniques. In battery systems, KF is used to estimate SOC by continuously updating battery state information using voltage and current measurements.

The Kalman Filter operates in two major stages: prediction and correction. In the prediction stage, the battery SOC is estimated using a mathematical battery model. In the correction stage, the predicted SOC is compared with actual sensor measurements, and the estimation error is minimized using correction equations.

The Kalman Filter is highly effective in handling noisy sensor measurements and improving estimation stability. However, the basic KF is mainly suitable for linear systems, whereas lithium-ion battery behavior is strongly nonlinear.[7]

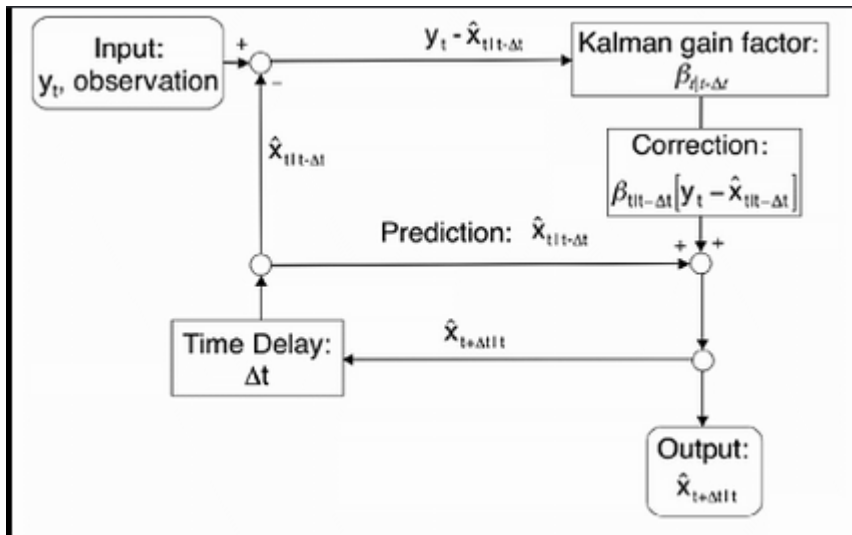


Figure 10: Prediction and Correction Process in Kalman Filter

Prediction Step :

In the prediction stage, the Kalman Filter estimates the next battery SOC value using the previous SOC estimate and the mathematical battery model.

Correction Step :

In the correction stage, the predicted SOC value is corrected using actual voltage and current measurements obtained from battery sensors. This correction process helps reduce estimation error and improve SOC accuracy.

MATHEMATICAL MODEL :

Prediction:

$$x(k) = A \cdot x(k-1) + B \cdot u(k)$$

Correction:

$$\hat{x}(k) = x(k) + K(z(k) - Hx(k))$$

Where:

- $x(k)$ = estimated state
- $u(k)$ = input
- K = Kalman gain
- $z(k)$ = measured output

Advantages

- Handles noisy sensor data
- Improves estimation accuracy
- Real-time estimation capability
- Reduces drift errors

Limitations

- Suitable mainly for linear systems
- Requires accurate battery model
- Limited nonlinear estimation capability

4.2 Extended Kalman Filter (EKF) :

The Extended Kalman Filter (EKF) is an advanced nonlinear extension of the basic Kalman Filter designed for nonlinear dynamic systems such as lithium-ion batteries. Since battery voltage and electrochemical behavior are nonlinear, EKF is widely used in Battery Management Systems for practical SOC estimation applications.

The EKF linearizes nonlinear battery equations around the current operating point and applies Kalman Filter estimation recursively. This enables improved SOC estimation accuracy under varying battery operating conditions such as temperature changes, dynamic load variation, and battery aging.

Due to its balance between estimation accuracy and computational complexity, EKF has become one of the most commonly used SOC estimation algorithms in electric vehicle Battery Management Systems.[7][8]

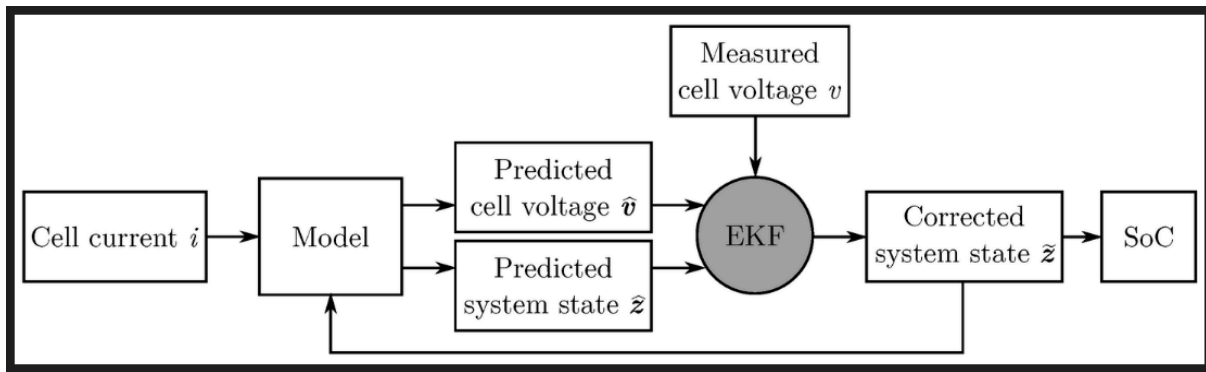


Figure 11: Extended Kalman Filter Based Battery SOC Estimation

Advantages

- Handles nonlinear battery systems
- Better accuracy than basic KF
- Real-time implementation possible
- Widely used in EV BMS

Limitations

- Requires linearization
- Approximation errors possible
- Complex tuning requirements

4.3 Unscented Kalman Filter (UKF)

The Unscented Kalman Filter (UKF) is an advanced nonlinear estimation algorithm developed to improve estimation accuracy for highly nonlinear systems. Unlike EKF, UKF does not rely on explicit linearization. Instead, it uses sigma points to model nonlinear state transitions more accurately.

UKF generally provides higher SOC estimation accuracy than EKF for nonlinear lithium-ion battery systems. The algorithm performs better under highly dynamic operating conditions and complex

nonlinear battery behavior caused by temperature variation, aging, and varying load conditions.

However, UKF requires higher computational power and memory resources, making embedded implementation more challenging for low-power Battery Management Systems.[8]

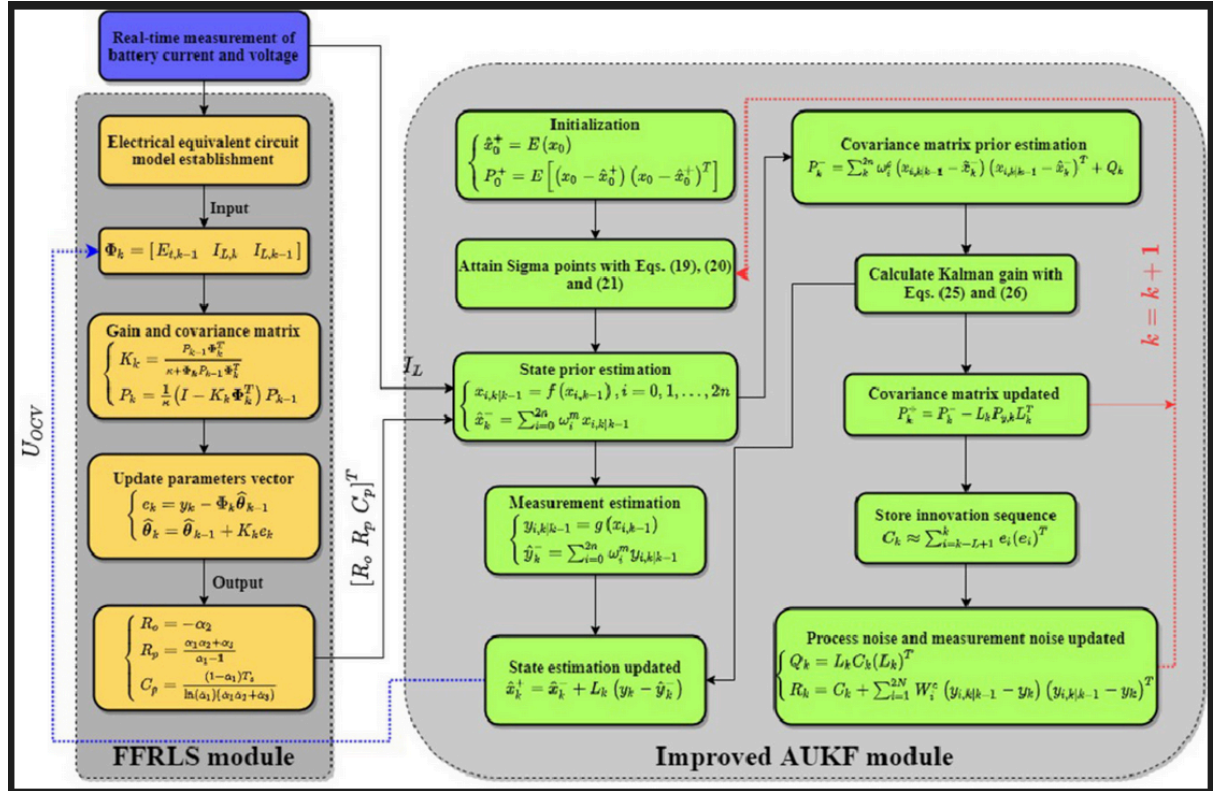


Figure 12: Unscented Kalman Filter Based SOC Estimation

Advantages

- Better nonlinear estimation accuracy
- Reduced approximation error
- Improved robustness

Limitations

- Higher computational complexity
- Increased memory requirement
- Difficult embedded implementation

4.4 KF vs EKF vs UKF Comparison

Different Kalman Filter algorithms provide different levels of estimation accuracy, nonlinear handling capability, and computational complexity. The selection of a suitable algorithm depends on battery behavior, hardware capability, estimation accuracy requirements, and real-time implementation constraints.

Parameter	KF	EKF	UKF
System Type	Linear	Nonlinear	Highly Nonlinear
Accuracy	Moderate	High	Very High
Nonlinear Handling	Poor	Good	Excellent
Computational Complexity	Low	Moderate	High
Real-Time Capability	High	High	Moderate
Memory Requirement	Low	Moderate	High
Embedded Suitability	Excellent	Good	Moderate
Main Limitation	Linear assumption	Approximation error	High computation

Table 3: Comparison of Kalman Filter Algorithms for SOC Estimation

Although Kalman Filter algorithms significantly improve SOC estimation accuracy compared to classical methods, their performance still depends on battery model quality and computational resources. These limitations encouraged researchers to explore machine learning and hybrid intelligent estimation approaches for improved robustness and adaptability.[7][8][9]

5. Machine Learning Approaches

Machine learning and deep learning techniques have gained significant attention for battery SOC and SOH estimation because of their ability to model highly nonlinear battery behavior without requiring explicit mathematical battery models. Unlike classical estimation methods, machine learning approaches learn complex relationships directly from battery datasets using historical operational data such as voltage, current, temperature, internal resistance, and charging/discharging patterns.

Modern lithium-ion batteries exhibit nonlinear and dynamic behavior influenced by temperature variation, battery chemistry, load conditions, and aging effects. Machine learning models can effectively capture these nonlinear patterns and improve estimation accuracy under real-world operating conditions.

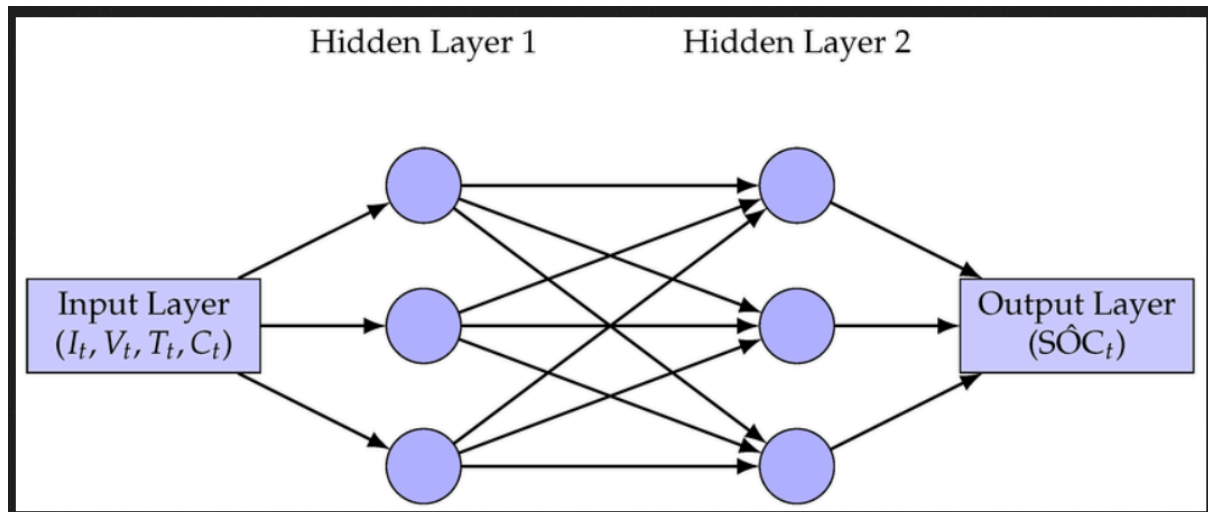
Machine learning approaches are increasingly used in electric vehicles, energy storage systems, and intelligent Battery Management Systems due to their adaptability, data-driven learning capability, and improved prediction performance. Common machine learning approaches used for SOC and SOH estimation include Artificial Neural Networks (ANN), Long Short-Term Memory (LSTM) networks, Gated Recurrent Units (GRU), Convolutional Neural Networks (CNN), Random Forest algorithms, and Support Vector Machines (SVM).[9][10]

5.1 Artificial Neural Network (ANN)

Artificial Neural Networks (ANNs) are data-driven machine learning models inspired by the structure and functioning of the human brain. ANNs learn nonlinear relationships between battery input parameters and SOC values through training processes using large battery datasets.

In SOC estimation, input parameters such as voltage, current, temperature, and charging/discharging cycles are fed into the neural network. The ANN processes these inputs through interconnected hidden layers and predicts the battery SOC at the output layer.

ANNs are capable of handling nonlinear battery behavior and achieving high estimation accuracy. However, ANN performance strongly depends on training data quality, network architecture, and computational resources.[10]



Here, I_t , V_t , T_t , and C_t represent battery current, voltage, temperature, and capacity-related input parameters respectively.

Figure 13: Artificial Neural Network Based SOC Estimation

Advantages

- Handles nonlinear battery behavior
- Good estimation accuracy
- Learns directly from data

Limitations

- Requires large training datasets
- Computationally expensive
- Black-box behavior

5.2 Long Short-Term Memory (LSTM) 🟢 🧠

Long Short-Term Memory (LSTM) networks are advanced recurrent neural networks specifically designed for sequential and time-series data analysis. Since battery behavior changes continuously over time, LSTM networks are highly suitable for SOC estimation applications.

LSTM networks contain memory cells capable of remembering previous battery states and learning long-term temporal dependencies from charging and discharging sequences. This allows LSTM models to capture dynamic battery behavior more effectively than conventional neural networks.

Due to their strong capability in handling sequential battery data, LSTM models have become one of the most widely researched deep learning methods for SOC and SOH estimation in lithium-ion batteries.[10][11]

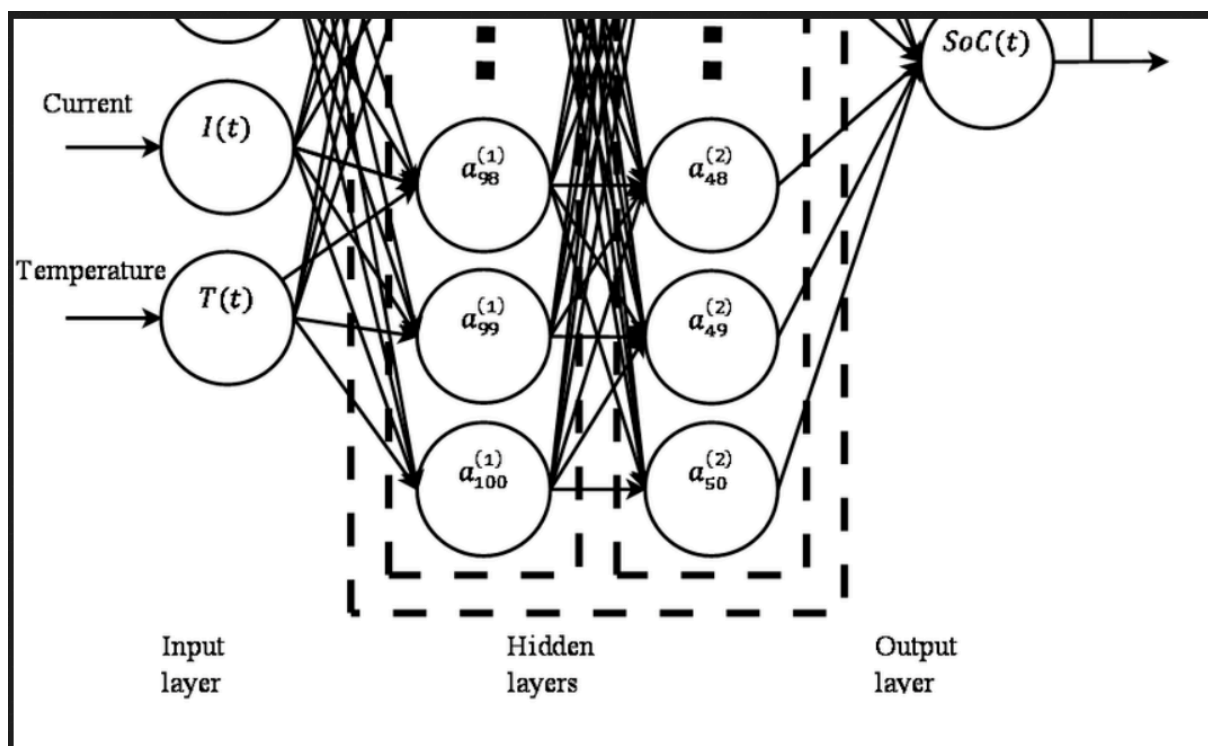


Figure 14: LSTM Network Architecture for Battery SOC Estimation

Advantages

- Excellent for time-series battery data
- High estimation accuracy
- Captures long-term battery behavior

Limitations

- High computational complexity
- Requires significant training time
- Memory intensive

5.3 Gated Recurrent Unit (GRU)

The Gated Recurrent Unit (GRU) is a simplified recurrent neural network architecture developed to reduce the computational complexity of LSTM models while maintaining effective sequential learning capability.

GRU models use gating mechanisms to capture temporal dependencies in battery charging and discharging data. Compared to LSTM networks, GRU models require fewer parameters and lower computational resources, making them more suitable for embedded and lightweight SOC estimation applications.[11]

Advantages

- Lower computational complexity
- Faster training
- Good sequential learning capability

Limitations

- Slightly lower learning capability than LSTM
- Accuracy depends on dataset quality

5.4 Convolutional Neural Network (CNN)

Convolutional Neural Networks (CNNs) are deep learning models primarily designed for automatic feature extraction and pattern recognition. In battery systems, CNNs are used to identify important patterns and hidden features from voltage, current, temperature, and degradation data.

CNN-based models can automatically extract relevant battery characteristics and improve SOC and SOH prediction accuracy by learning spatial and temporal patterns from battery datasets.[10]

Advantages

- Automatic feature extraction
- Good pattern recognition capability
- Improved prediction performance

Limitations

- Computationally intensive
- Requires large datasets

5.5 Random Forest 🌳⚡

Random Forest is an ensemble machine learning algorithm that combines multiple decision trees to improve prediction accuracy and robustness. In SOC estimation, multiple decision trees independently estimate SOC values, and the final output is obtained by averaging the predictions.

Random Forest algorithms are robust against noisy battery data and can handle nonlinear relationships effectively. However, large Random Forest models may increase memory usage and computational requirements.[10]

Advantages

- Good robustness
- Handles nonlinear data
- Resistant to overfitting

Limitations

- Larger memory requirement
- Reduced embedded suitability

5.6 Support Vector Machine (SVM)

Support Vector Machines (SVMs) are supervised machine learning algorithms used for regression and classification problems. SVM-based SOC estimation models identify optimal decision boundaries and relationships between battery parameters and SOC values.

SVM algorithms provide good estimation accuracy for smaller and well-structured datasets. However, training complexity increases significantly for large-scale battery datasets.[10]

Advantages

- Good accuracy for small datasets
- Effective nonlinear mapping

Limitations

- Difficult for large datasets
- High training complexity

5.7 ML Workflow for SOC/SOH Estimation

Machine learning-based SOC estimation typically follows a systematic workflow involving battery data collection, preprocessing, feature extraction, model training, validation, and prediction stages. Voltage, current, temperature, and degradation data are first collected from battery experiments and datasets. Relevant features are then extracted and used for training machine learning models.

After training, the model is validated using testing datasets to evaluate estimation accuracy and generalization capability. Finally, the trained model predicts battery SOC or SOH under real operating conditions.[11]

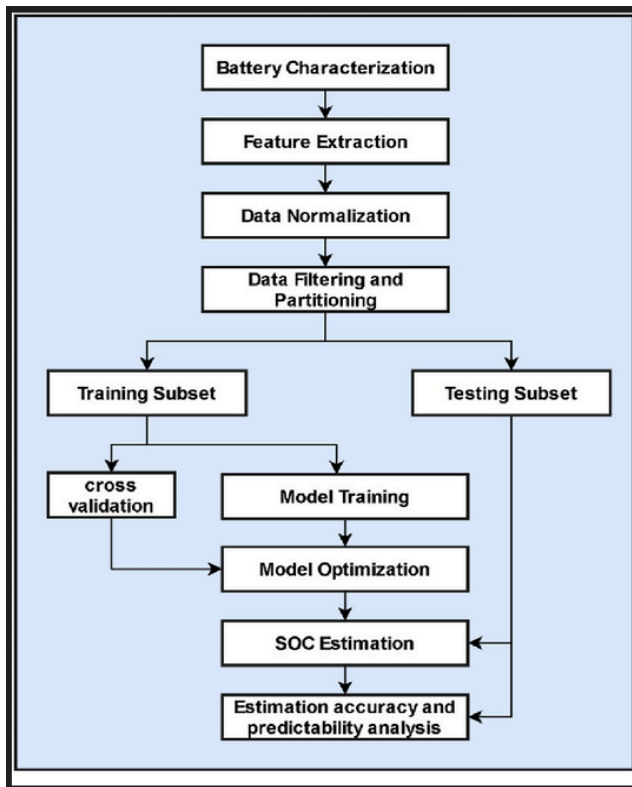


Figure 15: Machine Learning Workflow for Battery SOC and SOH Estimation

5.8 Dataset Requirements for ML Models

Required Parameter	Importance in ML-Based SOC Estimation
Voltage	Battery state indication
Current	Charge/discharge behavior
Temperature	Thermal behavior analysis
SOC Labels	Supervised learning target
SOH Labels	Aging estimation
Cycle Count	Degradation tracking

Table 4: Important Parameters for ML-Based Battery Estimation

5.9 Accuracy Comparison of ML Models

Model	Accuracy	Complexity	Time-Series Capability	Embedded Suitability
ANN	High	Moderate	Moderate	Moderate
LSTM	Very High	High	Excellent	Moderate
GRU	High	Moderate	Excellent	Good
CNN	High	High	Moderate	Moderate
Random Forest	Moderate	Moderate	Low	Moderate
SVM	Moderate	Moderate	Low	Good

Table 5: Comparison of Machine Learning Models for SOC Estimation

Machine learning approaches significantly improve SOC and SOH estimation accuracy by learning complex nonlinear battery behavior directly from data. However, these methods require large training datasets, computational resources, and efficient model optimization techniques for practical real-time Battery Management System implementation.[9][10][11]

6. Hybrid Estimation Techniques

Hybrid battery estimation techniques combine multiple estimation methods such as physics-based models, Kalman Filters, and machine learning algorithms to improve SOC and SOH estimation accuracy, robustness, and adaptability under real-world operating conditions.

Individual estimation techniques often suffer from specific limitations. Classical methods experience sensor drift and cumulative errors, Kalman Filters depend heavily on accurate battery models, and machine learning approaches require large training datasets and high computational resources. Hybrid estimation

approaches attempt to combine the strengths of different methods while minimizing their individual limitations.

Hybrid methods have become an important research direction in intelligent Battery Management Systems because lithium-ion battery behavior is highly nonlinear, chemistry-dependent, and dynamic. These methods provide improved estimation performance under varying temperature conditions, battery aging, load fluctuations, and real-world operating environments.[11][12]

6.1 Physics-Based Models

Physics-based battery models estimate SOC and SOH using electrochemical equations, equivalent circuit models, and battery dynamics. These models describe internal battery behavior such as ion movement, voltage response, internal resistance variation, and thermal characteristics.

Physics-based methods provide strong interpretability and physically meaningful predictions. However, accurate modeling of lithium-ion battery electrochemistry is highly complex and computationally intensive for real-time embedded implementation.[12]

Advantages

- Physically interpretable
- Good theoretical accuracy
- Captures electrochemical behavior

Limitations

- High mathematical complexity
- Difficult real-time implementation
- Requires accurate battery parameters

6.2 Kalman Filter + ML Fusion

Modern hybrid estimation systems often combine Kalman Filter algorithms with machine learning models to improve SOC estimation robustness and accuracy. In such systems, Kalman Filters handle real-time state estimation and noise reduction, while machine learning models learn nonlinear battery behavior and compensate for modeling errors.

The fusion of Kalman Filters and machine learning enables adaptive estimation under varying battery operating conditions and improves prediction stability in practical Battery Management Systems.[12]

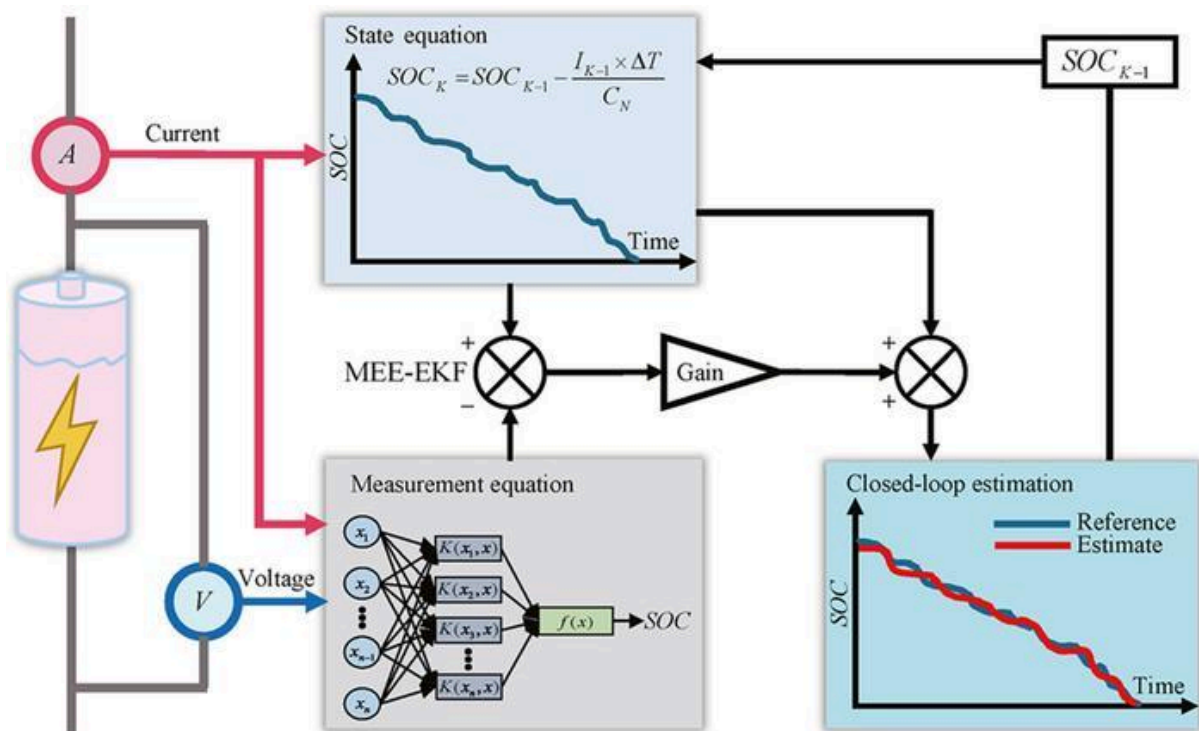


Figure 16: Hybrid SOC Estimation Architecture Combining Kalman Filter and Machine Learning

6.3 Adaptive Estimation Methods

Adaptive estimation methods dynamically update model parameters and estimation strategies according to changing battery conditions such as temperature variation, battery aging, and load changes. These methods improve estimation flexibility and robustness in real-time battery applications.

6.4 Physics-Informed Machine Learning

Physics-Informed Machine Learning combines physical battery knowledge with data-driven machine learning techniques to improve estimation reliability and generalization capability. Instead of relying purely on black-box learning, physics-informed models incorporate battery constraints, electrochemical relationships, and voltage behavior into the learning process.

Physics-informed approaches improve model interpretability, reduce unrealistic predictions, and enhance estimation accuracy under varying operating conditions. These methods are increasingly being explored in advanced battery intelligence and next-generation Battery Management Systems.[12][13]

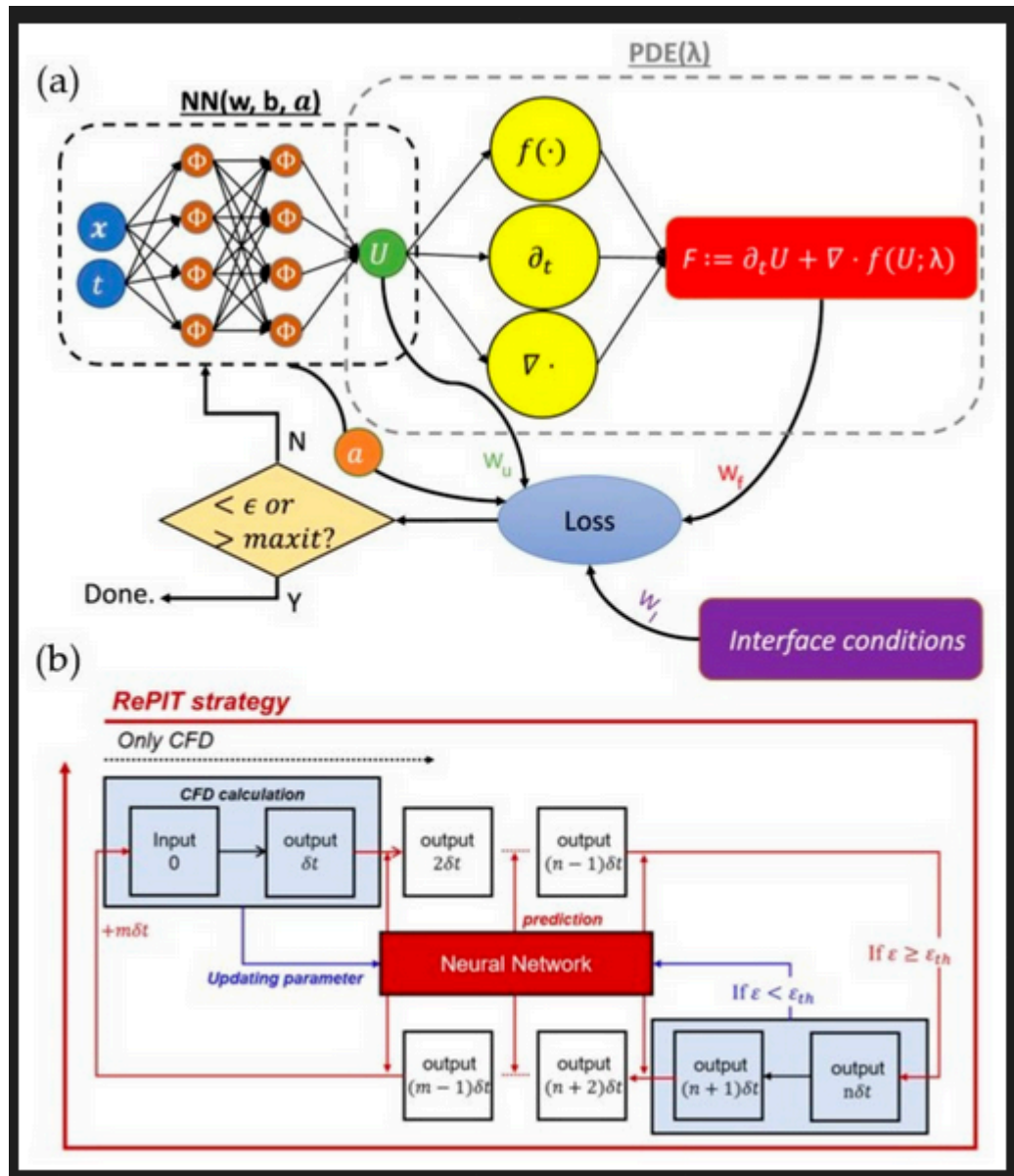


Figure 17: Physics-Informed Neural Networks for Advanced Thermal Management in Electronics and Battery Systems

6.5 Hybrid Architecture Diagram

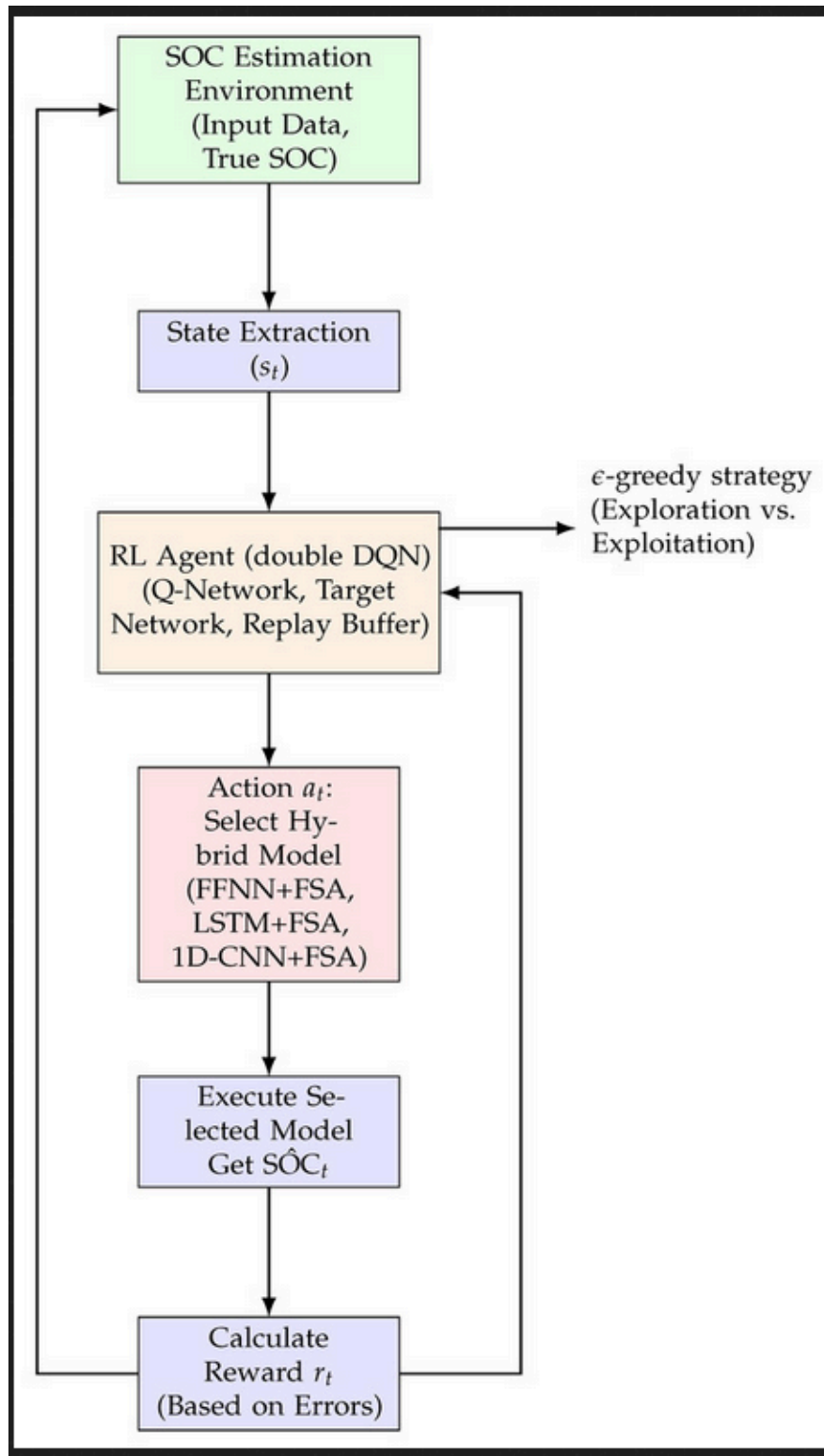


Figure 18: General Hybrid Battery SOC Estimation Framework

Advantages

- Improved estimation accuracy
- Better nonlinear handling
- Increased robustness
- Adaptive estimation capability
- Improved real-time performance

Challenges

- Higher computational complexity
- Increased implementation difficulty
- Large training data requirement
- Embedded deployment challenges

6.8 Real-World Applications 🚗

Hybrid SOC estimation methods are increasingly used in electric vehicles, renewable energy storage systems, aerospace battery systems, and smart Battery Management Systems. These methods improve operational safety, charging optimization, battery lifespan, and real-time monitoring capability under practical operating conditions.

Hybrid estimation techniques represent an important future direction for intelligent battery management because they combine the strengths of physics-based modeling, state estimation algorithms, and machine learning approaches. These methods are highly suitable for next-generation Edge-AI and TinyML-enabled Battery Management Systems.[11][12][13]

7. Comparative Analysis of Estimation Methods 📊

Different SOC and SOH estimation methods exhibit varying levels of estimation accuracy, computational complexity, hardware requirements, and real-time implementation capability. Classical estimation methods are simple and computationally efficient but suffer from lower long-term accuracy. Kalman Filter algorithms improve estimation stability and noise handling, while machine learning and hybrid approaches provide better nonlinear modeling and adaptive learning capability.

The selection of a suitable estimation technique depends on application requirements such as estimation accuracy, computational power, battery chemistry, embedded hardware capability, and real-time operating constraints.[13]

Method	Accuracy	Complexity	Real-Time Capability	Hardware Requirement	Advantage	Limitations
Coulomb Counting	Moderate	Low	Excellent	Low	Simple and fast	Error accumulation
OCV Method	Good at rest	Low	Limited	Low	Accurate under resting condition	Requires resting time
KF	Moderate	Low	Excellent	Moderate	Noise handling	Linear assumption
EKF	High	Moderate	Excellent	Moderate	Nonlinear estimation	Approximation error
UKF	Very High	High	Good	High	Better nonlinear handling	High computation
ANN	High	Moderate	Moderate	Moderate	Learns nonlinear behavior	Needs large dataset
LSTM	Very High	High	Moderate	High	Excellent time-series learning	Computationally expensive
Hybrid Methods	Excellent	Very High	Moderate	Very High	Highest robustness	Complex implementation

Table 6: Comparative Analysis of SOC Estimation Methods

Comparative analysis shows that no single estimation method is universally optimal for all battery applications. Classical methods remain useful for low-complexity systems, while Kalman Filter algorithms provide improved estimation robustness. Machine learning and hybrid approaches offer superior nonlinear modeling capability and estimation accuracy but require higher computational resources and dataset quality. Therefore, modern research increasingly focuses on lightweight hybrid intelligent estimation methods suitable for Edge-AI and TinyML-enabled Battery Management Systems.[10][11][12][13]

8. Research Challenges

Despite significant advancements in battery SOC and SOH estimation technologies, several technical and practical challenges still limit estimation accuracy, reliability, and real-time deployment in Battery Management Systems. Lithium-ion batteries exhibit highly nonlinear, dynamic, and chemistry-dependent behavior, making accurate state estimation a complex engineering problem.

Battery operating conditions continuously change due to charging/discharging cycles, temperature variation, aging effects, load fluctuations, and environmental factors. In addition, embedded hardware limitations and noisy sensor measurements further increase estimation complexity in practical applications.[12][13]

8.1 Nonlinear Battery Behavior

Lithium-ion batteries exhibit highly nonlinear voltage, current, and electrochemical behavior during charging and discharging operations. Battery response varies dynamically with SOC, temperature, load conditions, and operating history. These nonlinear characteristics reduce the effectiveness of simple estimation algorithms and increase the complexity of accurate SOC prediction.

8.2 Battery Chemistry Dependence

Different lithium-ion battery chemistries such as LFP, NMC, and LCO exhibit different voltage-SOC relationships, thermal characteristics, degradation behavior, and internal resistance variation. As a result, estimation models developed for one battery chemistry may not perform accurately for another chemistry, creating major challenges for universal SOC estimation systems.

8.3 Aging Effects

Battery characteristics continuously degrade over time due to repeated charging/discharging cycles, electrochemical side reactions, and thermal stress. Aging causes capacity fade, increased internal resistance, and altered voltage behavior, reducing the accuracy of estimation models trained using fresh battery data.

8.4 Temperature Sensitivity

Temperature significantly affects lithium-ion battery performance, electrochemical reactions, internal resistance, and voltage response. Extreme temperatures may cause inaccurate SOC estimation, reduced battery efficiency, and safety risks such as thermal runaway. Therefore, temperature-aware estimation models are essential for reliable Battery Management Systems.

8.5 Sensor Noise and Drift

Voltage, current, and temperature sensors used in Battery Management Systems are affected by noise, calibration errors, and sensor drift during long-term operation. Small sensor measurement errors accumulate over time and significantly reduce estimation accuracy, especially in current integration-based methods such as Coulomb Counting.

8.6 Dataset Limitations

Machine learning-based estimation methods require large high-quality battery datasets containing voltage, current, temperature, SOC, and SOH information under diverse operating conditions. However, publicly available battery datasets often contain missing values, noisy measurements, limited chemistry diversity, and insufficient real-world operating scenarios, reducing model generalization capability.

8.7 Embedded System Constraints

Advanced estimation algorithms such as deep learning models and hybrid intelligent systems require significant computational resources, memory capacity, and power consumption. However, Battery Management Systems typically operate on resource-constrained embedded hardware and

microcontrollers, making real-time deployment of complex estimation models challenging.

Addressing these research challenges is essential for developing accurate, reliable, and lightweight SOC estimation systems suitable for next-generation electric vehicles and intelligent Battery Management Systems. Recent research increasingly focuses on hybrid estimation techniques, physics-informed machine learning, TinyML, and Edge-AI solutions to overcome these limitations.[12][13][14]

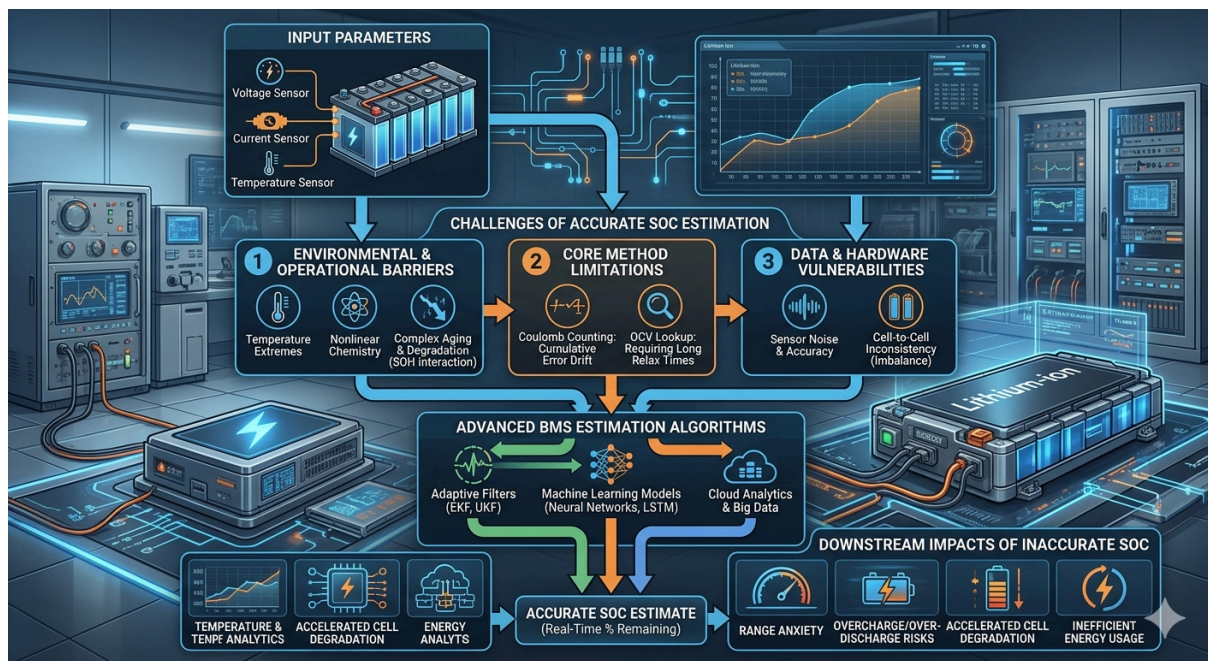


Figure 19: Major Challenges in Battery SOC and SOH Estimation(ai generated)

9. Edge-AI and TinyML Possibilities

The rapid growth of intelligent electric vehicles, portable electronics, and smart energy storage systems has increased the demand for lightweight and real-time battery intelligence solutions. Traditional cloud-based machine learning systems often require high computational resources, internet connectivity, and significant power consumption, making them unsuitable for embedded Battery Management Systems.

Recent advancements in Edge-AI and TinyML technologies have enabled the deployment of lightweight machine learning models directly on embedded

battery hardware and microcontrollers. These technologies allow real-time SOC and SOH estimation with reduced latency, lower power consumption, and improved operational reliability.[13][14]

9.1 TinyML for Battery Management Systems

TinyML refers to the deployment of lightweight machine learning models on low-power embedded systems and microcontrollers. In Battery Management Systems, TinyML enables intelligent SOC and SOH estimation directly on embedded hardware without requiring cloud-based processing.

TinyML-based estimation systems improve real-time decision-making capability while reducing communication delay, power consumption, and hardware cost. These systems are highly suitable for electric vehicles, portable devices, and edge battery applications.

9.2 Edge-AI Based SOC Estimation

Edge-AI refers to artificial intelligence processing performed directly on local hardware devices instead of centralized cloud servers. In battery systems, Edge-AI enables local SOC estimation, anomaly detection, thermal monitoring, and battery health prediction directly inside the Battery Management System.

Edge-AI-based SOC estimation improves operational safety, reduces latency, enhances data privacy, and enables real-time intelligent battery monitoring under dynamic operating conditions.

9.3 Embedded Hardware Constraints

Although machine learning methods provide improved estimation accuracy, embedded Battery Management Systems typically operate under severe hardware limitations such as low memory capacity, limited processing power, low energy availability, and strict real-time operating requirements.

Therefore, lightweight estimation algorithms with reduced computational complexity are essential for practical embedded implementation.

9.4 Lightweight ML Models

Lightweight machine learning models such as compressed neural networks, optimized GRU models, lightweight LSTM architectures, and quantized neural

networks are increasingly being developed for TinyML-based battery estimation applications.

These optimized models reduce memory usage and computational complexity while maintaining acceptable SOC estimation accuracy for embedded deployment.

9.5 Real-Time TinyML Inference 🚗

Real-time TinyML inference enables continuous SOC and SOH prediction directly on Battery Management System hardware during battery operation. Real-time estimation improves charging optimization, thermal management, fault detection, battery lifespan, and operational safety in electric vehicles and smart battery systems.

9.6 Future Research Directions 🔬

Future battery intelligence research is expected to focus on physics-informed TinyML models, adaptive hybrid estimation systems, federated battery learning, multi-chemistry SOC estimation, and ultra-low-power embedded AI hardware for next-generation Battery Management Systems.

Combining machine learning, battery physics, and embedded intelligence is expected to play a critical role in the development of future intelligent electric mobility and energy storage technologies.

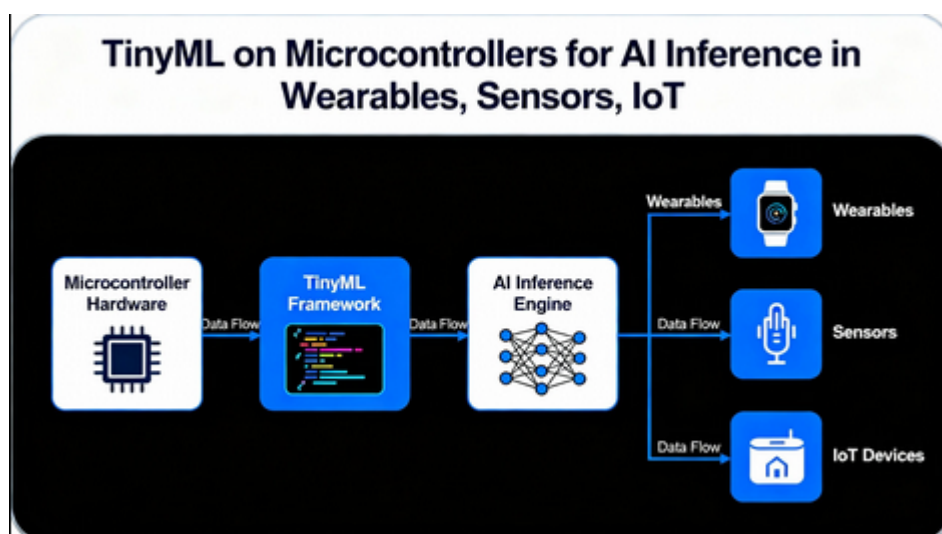


Figure 20: TinyML and Edge-AI Based Intelligent electronic System

TinyML and Edge-AI technologies represent promising future directions for intelligent battery estimation systems because they enable lightweight, real-time, and energy-efficient deployment of machine learning models directly on embedded Battery Management Systems. These technologies are expected to significantly improve battery monitoring, safety, efficiency, and autonomous energy management in future electric vehicles and smart energy systems.[13][14][15]

10. Conclusion

Accurate estimation of battery State of Charge (SOC) and State of Health (SOH) is essential for ensuring the safety, reliability, efficiency, and lifespan of lithium-ion battery systems used in electric vehicles, portable electronics, renewable energy storage systems, and smart energy applications. Since SOC and SOH cannot be directly measured using physical sensors, intelligent estimation techniques are required for effective Battery Management System operation.

This report presented a comprehensive study of various SOC and SOH estimation methods, including classical techniques such as Coulomb Counting and Open Circuit Voltage estimation, advanced Kalman Filter algorithms, machine learning approaches, and modern hybrid intelligent estimation techniques. Classical methods provide simplicity and real-time capability but suffer from estimation drift and limited dynamic performance. Kalman Filter algorithms improve estimation stability and noise handling, while machine learning methods provide superior nonlinear learning capability and adaptive prediction performance.

Hybrid estimation techniques combining physics-based modeling, Kalman Filters, and machine learning approaches represent an important modern research direction for improving estimation robustness and accuracy under practical operating conditions. In addition, emerging TinyML and Edge-AI technologies provide promising opportunities for implementing lightweight intelligent estimation systems directly on embedded Battery Management System hardware.

Despite significant progress in battery intelligence research, several challenges such as nonlinear battery behavior, chemistry dependence, aging effects, sensor noise, dataset limitations, and embedded hardware constraints continue to affect estimation performance. Future research is expected to focus on physics-informed machine learning, adaptive hybrid estimation systems, multi-chemistry SOC estimation, and ultra-lightweight embedded AI solutions for next-generation intelligent battery systems.

Overall, intelligent SOC and SOH estimation technologies will play a critical role in the future development of electric mobility, smart energy storage, and advanced Battery Management Systems.

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Thank you.